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SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

SOFTWARE: CISCO PACKET TRACER, WIRESHARK

83%

EXPERIMENTS

QUESTIONS:4

Total Marks:40

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.
2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.
3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the

Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.

4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.
5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding ; limited or partial integration	Poor understanding ; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate	Good analysis with	Basic analysis with	Poor or incorrect

		interpretation and validation of results	reasonable interpretation	limited insights	analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

	Understanding of Concepts (25%)	Experimental Design (30%)	Use of Tools (30%)	Analysis of Results (15%)	Documentation (10%)	
Q1	2	2	2	1.5	1	8.5
Q2	2	2	1.5	1	1	7.5
Q3	2.5	3	2	1.5	0.5	9.5
Q4	2	2	1.5	1	0.5	7
Q5	1.5	3	2	1.5	1	9
						41.5

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1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.

LAN Using Switch and Ethernet Frame Analysis

Aim

To design a simple LAN with two PCs connected through a switch, configure IP addresses, verify connectivity using ping, and analyze Ethernet frame fields using Wireshark.

Network Topology

PC0 ----- Switch ----- PC1

Device	IP Address	Subnet Mask
PC0	192.168.1.10	255.255.255.0
PC1	192.168.1.20	255.255.255.0

Both PCs are in the same network: **192.168.1.0/24**.

Procedure

1. Open Cisco Packet Tracer.

2. Add two PCs and one switch.

3. Connect both PCs to the switch using Copper Straight-Through cables.

4. Configure the IP address of PC0 as **192.168.1.10**.

5. Configure the IP address of PC1 as **192.168.1.20**.

6. Set the subnet mask as **255.255.255.0** for both PCs.
7. Open the Command Prompt on PC0.
8. Execute

```
ping 192.168.1.20
```

9. Verify that Echo Reply messages are received.
10. Capture the communication using Wireshark and examine the Ethernet frame.

Ethernet Frame Fields

1. Source MAC Address

It identifies the MAC address of the device sending the frame.

2. Destination MAC Address

It identifies the MAC address of the device receiving the frame.

3. EtherType

It identifies the protocol carried inside the Ethernet frame, such as IPv4 or ARP.

4. Frame Check Sequence (FCS)

FCS is used for error detection. The receiver checks the FCS to determine whether the frame was damaged during transmission.

Explanation

The Data Link Layer uses MAC addresses to deliver frames within a local network. The source MAC address identifies the sender, while the destination MAC address identifies the receiver. The switch uses MAC addresses to forward the frame to the correct destination port.

Result

The two PCs successfully communicated through the switch using the ping command. The Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and

FCS were identified successfully.

Experiment 1: Two PCs Connected through Switch

PC0

192.168.1.10

Switch0

2960-24TT

PC1

192.168.1.20

PC0

Physical Config Desktop Programming Attributes

Command Prompt

Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.20:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>

RESULT:

Ping is successful between PC0 and PC1.
This confirms that both PCs are in the same network and can communicate through the switch.

Capturing from FastEthernet0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl+/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	PC0	PC1	ICMP	98	Echo (ping) request id=0x0001, seq=1/256,
2	0.000214	PC1	PC0	ICMP	98	Echo (ping) reply id=0x0001, seq=1/256,
3	1.001266	PC0	PC1	ICMP	98	Echo (ping) request id=0x0001, seq=2/512,
4	1.001476	PC1	PC0	ICMP	98	Echo (ping) reply id=0x0001, seq=2/512,
5	2.002400	PC0	PC1	ICMP	98	Echo (ping) request id=0x0001, seq=3/768,
6	2.002613	PC1	PC0	ICMP	98	Echo (ping) reply id=0x0001, seq=3/768,
7	3.003450	PC0	PC1	ICMP	98	Echo (ping) request id=0x0001, seq=4/1024,
8	3.003655	PC1	PC0	ICMP	98	Echo (ping) reply id=0x0001, seq=4/1024,
9	4.004362	PC0	PC1	ICMP	98	Echo (ping) request id=0x0001, seq=5/1280,
10	4.004572	PC1	PC0	ICMP	98	Echo (ping) reply id=0x0001, seq=5/1280,
11	5.005365	PC0	PC1	ICMP	98	Echo (ping) request id=0x0001, seq=6/1536,
12	5.005572	PC1	PC0	ICMP	98	Echo (ping) reply id=0x0001, seq=6/1536,

> Frame 1: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface \Device\NPF_{3E6B7A0B-1} Ethernet II, Src: 00:50:79:66:68:01 (00:50:79:66:68:01), Dst: 00:50:79:66:68:02 (00:50:79:66:68:02)
> Destination: 00:50:79:66:68:02 (00:50:79:66:68:02)
> Source: 00:50:79:66:68:01 (00:50:79:66:68:01)
> Type: IPv4 (0x0800)
> Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.1.20
> Internet Control Message Protocol

Destination MAC AddressSource MAC AddressEtherType (0x0800)FCS (Frame Check Sequence)

0000 00 50 79 66 68 02 00 50 79 66 68 01 08 00 45 00 ...Pyfh...Pyfh...E.
0010 00 54 00 01 00 00 40 01 c7 9c c0 a8 01 0a c0 a8 .T.....
0020 01 14 08 00 34 6a 00 01 00 01 61 62 63 64 65 664;.....abcdef
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmn opqrstuv
0040 77 78 79 7a wxyz

5a 3d [Z =]
FCS (4 Bytes)

2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.

Two LANs Connected Through a Router and IPv4 Analysis

Aim

To create two LANs connected through a router, configure IPv4 addresses and default gateways, verify communication, and analyze IPv4 header fields.

Network Topology:

PC0 ---- Switch0 ---- Router ---- Switch1 ---- PC1

IP Address Configuration

LAN 1

Device	IP Address	Subnet Mask	Gateway
PC0	192.168.1.10	255.255.255.0	192.168.1.1
Router Interface	192.168.1.1	255.255.255.0	-

LAN 2

Device	IP Address	Subnet Mask	Gateway
PC1	192.168.2.10	255.255.255.0	192.168.2.1
Router Interface	192.168.2.1	255.255.255.0	-

Procedure

1. Open Cisco Packet Tracer.
2. Add two PCs, two switches, and one router.
3. Connect the devices correctly.
4. Configure IP addresses for PC0 and PC1.
5. Configure the router interfaces.
6. Enable the router interfaces using:

Explanation

The Network Layer is responsible for logical addressing and routing packets between different networks. The router receives the packet, checks the destination IP address, and forwards the packet toward the destination network.

Result

Successful communication was established between the two different LANs. The IPv4 header fields were captured and analyzed successfully.

Question 2: Two LANs Connected Through a Router

PC0

192.168.1.10

/24

Switch0

Router0

Switch1

PC1

192.168.2.10

/24

LAN 1: 192.168.1.0/24

LAN 2: 192.168.2.0/24

LAN 1 - Configuration

PC0 IP Address : 192.168.1.10

Subnet Mask : 255.255.255.0

Default Gateway : 192.168.1.1

LAN 2 - Configuration

PC1 IP Address : 192.168.2.10

Subnet Mask : 255.255.255.0

Default Gateway : 192.168.2.1

Router Configuration

Fa0/0 : 192.168.1.1 /24

Fa0/1 : 192.168.2.1 /24

PC0

Physical Config Desktop Programming Attributes

Command Prompt

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>

Result:

End-to-end communication is successful between PC0 (192.168.1.10) and PC1 (192.168.2.10) through the router.

Capturing from Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl+>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.10	192.168.2.10	ICMP	98	Echo (ping) request id=0x0001
2	0.000987	192.168.2.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0x0001
3	1.002134	192.168.1.10	192.168.2.10	ICMP	98	Echo (ping) request id=0x0002
4	1.003021	192.168.2.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0x0002
5	2.005124	192.168.1.10	192.168.2.10	ICMP	98	Echo (ping) request id=0x0003
6	2.006012	192.168.2.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0x0003

> Frame 1: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface \Device\NPF_{38S...}

> Ethernet II, Src: 00:50:79:66:68:01, Dst: 00:50:79:66:68:02

> Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.2.10

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 84

Identification: 0x1c46 (7238)

> Flags: 0x00

Time to Live: 128

Protocol: ICMP (1)

Header Checksum: 0x9a12 [correct]

Source Address: 192.168.1.10

Destination Address: 192.168.2.10

> Internet Control Message Protocol

0000 00 50 79 66 68 02 00 50 79 66 68 01 08 00 45 00 ..Pyfh...Pyfh...E.

0010 00 54 1c 46 00 00 80 01 9a 12 c0 a8 01 0a c0 a8 ...T.F.....

0020 02 0a 08 00 f7 ff 00 01 00 01 61 62 63 64 65 66abctdefgh.

0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ijklbmnoqrstu.

0040 77 78 79 7a wxyz

IPv4 Header Fields

Source IP Address

Destination IP Address

TTL (Time To Live)

Protocol Number

Header Length

Role of Network Layer:

• Network layer uses logical IP addresses.

• Router forwards packets between different networks.

• TTL prevents infinite looping of packets.

• Protocol field identifies the upper layer (ICMP in this case).

3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.

ARP Request and ARP Reply Analysis

Aim

To construct a LAN with multiple hosts, clear the ARP cache, initiate communication, capture ARP Request and ARP Reply packets, and analyze how ARP converts IP addresses into MAC addresses.

Network Topology

PC0 -----\
Switch
PC1 -----/

IP Address Configuration

Device	IP Address	Subnet Mask
PC0	192.168.1.10	255.255.255.0
PC1	192.168.1.20	255.255.255.0

Procedure

1. Open Cisco Packet Tracer.
2. Add multiple PCs and connect them through a switch.
3. Configure appropriate IP addresses.
4. Clear the ARP cache on PC0.
5. Send a ping from PC0 to PC1:
6. Capture the packets using Wireshark.
7. Identify the ARP Request packet.
8. Identify the ARP Reply packet.
9. Examine the sender and target IP and MAC addresses.

ARP Request

When PC0 knows the destination IP address but does not know the MAC address, it sends an ARP Request.

Example:

Sender IP Address: 192.168.1.10

Sender MAC Address: PC0 MAC Address

Target IP Address: 192.168.1.20

Target MAC Address: Unknown

Explanation

ARP stands for **Address Resolution Protocol**. It is used to find the physical MAC address associated with an IPv4 address in a local network.

PC needs MAC address

↓

ARP Request is broadcast

↓

Destination receives request

↓

ARP Reply is sent

↓

MAC address is stored in ARP cache

↓

Data transmission begins

Result

The ARP Request and ARP Reply packets were successfully captured. ARP successfully resolved the destination IP address into the corresponding MAC address before data transmission.

Experiment 3: ARP Request and ARP Reply Analysis

PC0
192.168.1.10

Switch0
2960-24TT

PC1
192.168.1.20

Network: 192.168.1.0/24

PC0

Physical Config Desktop Programming Attributes

Command Prompt

C:\>arp -d
Deleting all ARP entries.

C:\>arp -a
Internet Address Physical Address Type
(no entries)

C:\>ping 192.168.1.20
Pinging 192.168.1.20 with 32 bytes of data:
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.20:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>

PC0 - ARP Cache (After Ping)

Internet Address	Physical Address	Type	Interface
192.168.1.20	00D0.BA3A.7C01	dynamic	FastEthernet0

RESULT:
ARP Request and ARP Reply packets were captured successfully.
ARP resolved the IP address (192.168.1.20) to the MAC address (00D0.BA3A.7C01) before data transmission.

ARP Process

PC0 wants to send data to PC1 (192.168.1.20) → PC0 broadcasts ARP Request (Who has 192.168.1.20?) → PC1 receives request and sends ARP Reply → PC0 receives reply and knows PC1 MAC address (00D0.BA3A.7C01) → Now data transmission (ICMP Echo Request/Reply) takes place

Capturing from FastEthernet0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl+>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Pc0	Broadcast	ARP	60	Who has 192.168.1.20? Tell 192.168.1.10
2	0.000296	Pc1	Pc0	ARP	60	192.168.1.20 is at 00D0.BA3A.7C01
3	0.001192	Pc0	Pc1	ICMP	98	Echo (ping) request id=0x0001, seq=1/256,
4	0.001414	Pc1	Pc0	ICMP	98	Echo (ping) reply id=0x0001, seq=1/256,
5	1.003260	Pc0	Pc1	ICMP	98	Echo (ping) request id=0x0001, seq=2/512,
6	1.003454	Pc1	Pc0	ICMP	98	Echo (ping) reply id=0x0001, seq=2/512,
7	2.005241	Pc0	Pc1	ICMP	98	Echo (ping) request id=0x0001, seq=3/768,
8	2.005449	Pc1	Pc0	ICMP	98	Echo (ping) reply id=0x0001, seq=3/768,

> Frame 1: 60 bytes on wire (480 bits), 60 bytes captured
> Ethernet II, Src: 00D0.BA3A.7C00, Dst: Broadcast (ff:ff:ff:ff:ff:ff)
▼ Address Resolution Protocol (request)
Hardware type: Ethernet (1)
Protocol type: IPv4 (0x0800)
Hardware size: 6
Protocol size: 4
Opcode: request (1)
Sender MAC address: 00D0.BA3A.7C00
Sender IP address: 192.168.1.10
Target MAC address: 00:00:00:00:00:00
Target IP address: 192.168.1.20
ARP Request

> Frame 2: 60 bytes on wire (480 bits), 60 bytes captured
> Ethernet II, Src: 00D0.BA3A.7C01, Dst: 00D0.BA3A.7C00
▼ Address Resolution Protocol (reply)
Hardware type: Ethernet (1)
Protocol type: IPv4 (0x0800)
Hardware size: 6
Protocol size: 4
Opcode: reply (2)
Sender MAC address: 00D0.BA3A.7C01
Sender IP address: 192.168.1.20
Target MAC address: 00D0.BA3A.7C00
Target IP address: 192.168.1.10
ARP Reply

0000	ff ff ff ff ff ff 00 d0 ba 3a 7c 00 08 06 00 01		0000	00 d0 ba 3a 7c 00 00 d0 ba 3a 7c 01 08 06 00 01	
0010	08 00 06 04 00 01 00 d0 ba 3a 7c 00 c0 a8 01 0a	..9...	0010	08 00 06 04 00 02 00 d0 ba 3a 7c 01 c0 a8 01 14	..91..
0020	00 00 00 00 00 00 c0 a8 01 14		0020	00 d0 ba 3a 7c 00 c0 a8 01 0a	
0030	: }		0030	: }	

4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

ICMP Echo Request and Echo Reply Analysis

Aim

To configure two PCs connected through a router, test connectivity using ping, capture ICMP traffic, analyze ICMP fields, and calculate Round Trip Time.

Network Topology

PC0 ----- Router ----- PC1

IP Address Configuration

Device	IP Address	Subnet Mask	Gateway
PC0	192.168.1.10	255.255.255.0	192.168.1.1
Router Interface 1	192.168.1.1	255.255.255.0	-
Router Interface 2	192.168.2.1	255.255.255.0	-
PC1	192.168.2.10	255.255.255.0	192.168.2.1

Procedure

1. Open Cisco Packet Tracer.
2. Add two PCs and one router.
3. Connect PC0 and PC1 through the router.
4. Configure the required IP addresses.

5. Enable the router interfaces.
6. Configure the default gateways.
7. From PC0, execute:

```
ping 192.168.2.10
```

8. Capture the ICMP traffic using Wireshark.
9. Identify ICMP Echo Request and Echo Reply packets.
10. Examine Type, Code, Identifier, and Sequence Number.

ICMP Echo Request

The sender sends an ICMP Echo Request to check whether the destination device is reachable.

Identifier

Used to identify the ping session.

Sequence Number

Used to identify and match individual Echo Request and Echo Reply packets.

Round Trip Time – RTT

RTT is the total time required for a packet to travel from the sender to the destination and return.

Explanation

ICMP is used for network troubleshooting and connectivity testing. The ping command uses ICMP Echo Request and Echo Reply messages to determine whether a destination is reachable and to measure communication delay.

Result

The ICMP Echo Request and Echo Reply messages were successfully captured and analyzed. The RTT was calculated successfully, confirming network connectivity.

Experiment 4: ICMP Echo Request and Echo Reply Analysis

PC0

192.168.1.10/24

Gateway: 192.168.1.1

G0/0

PC1

192.168.2.10/24

Gateway: 192.168.2.1

G0/1

PC0 is pinging PC1

Command Prompt

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time=2ms TTL=126

Reply from 192.168.2.10: bytes=32 time=1ms TTL=126

Reply from 192.168.2.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.10:

Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>

RESULT:

ICMP Echo Request and Echo Reply messages

were captured successfully.

The ping is successful and RTT is calculated.

Round Trip Time (RTT) Calculation

Request Time : 0.000000 s

Reply Time : 0.001245 s

RTT = 0.001245 - 0.000000 = 0.001245 s

RTT = 1.245 ms

Capturing from FastEthernet0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl+>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.10	192.168.2.10	ICMP	98	Echo (ping) request id=0x0001, seq=1/256,
2	0.001245	192.168.2.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0x0001, seq=1/256,
3	1.001250	192.168.1.10	192.168.2.10	ICMP	98	Echo (ping) request id=0x0001, seq=2/512,
4	1.002487	192.168.2.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0x0001, seq=2/512,
5	2.002493	192.168.1.10	192.168.2.10	ICMP	98	Echo (ping) request id=0x0001, seq=3/768,
6	2.003720	192.168.2.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0x0001, seq=3/768,

> Frame 1: 98 bytes on wire (784 bits), 98 bytes captured

> Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.2.10

> Internet Control Message Protocol

Type: 8 (Echo (ping) request)

Code: 0

Checksum: 0x5acb [correct]

[Checksum Status: Good]

Identifier (BE): 1 (0x0001)

Identifier (LE): 256 (0x0100)

Sequence Number (BE): 1 (0x0001)

Sequence Number (LE): 256 (0x0100)

[Response frame: 2]

Data (32 bytes)

ICMP Echo Request

> Frame 2: 98 bytes on wire (784 bits), 98 bytes captured

> Internet Protocol Version 4, Src: 192.168.2.10, Dst: 192.168.2.10

> Internet Control Message Protocol

Type: 0 (Echo (ping) reply)

Code: 0

Checksum: 0x5ccb [correct]

[Checksum Status: Good]

Identifier (BE): 1 (0x0001)

Identifier (LE): 256 (0x0100)

Sequence Number (BE): 1 (0x0001)

Sequence Number (LE): 256 (0x0100)

[Request frame: 1]

Data (32 bytes)

ICMP Echo Reply

Important ICMP Fields

Field	Echo Request (From PC0)	Echo Reply (From PC1)	Description
Type	8	0	8 = Echo Request, 0 = Echo Reply
Code	0	0	No error
Identifier	0x0001 (1)	0x0001 (1)	Used to identify the ping session
Sequence Number	1, 2, 3 ...	1, 2, 3 ...	Used to match request and reply packets

ICMP Communication Flow

PC0 (192.168.1.10)

PC1 (192.168.2.10)

Echo Request (Type = 8, Code = 0)

Echo Reply (Type = 0, Code = 0)

5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

Client-Server Network, TCP and HTTP Analysis

Aim

To create a client-server network, configure a web server and client PC, enable HTTP service, access the webpage, and analyze TCP three-way handshake and HTTP communication.

Network Topology

Client PC ----- Switch ----- Web Server

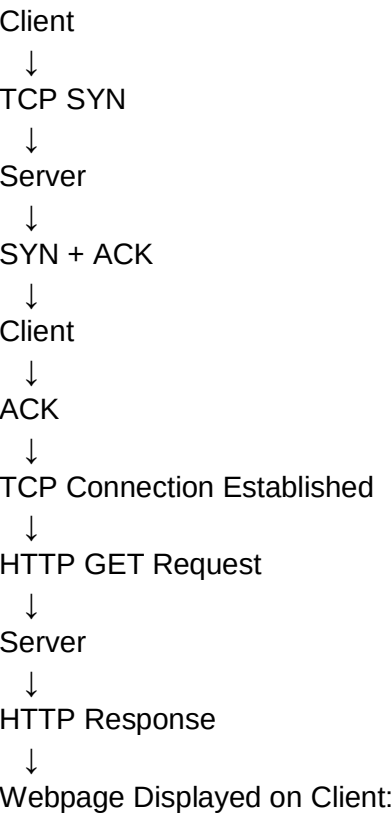
Procedure

- 1. Open Cisco Packet Tracer.
- 2. Add one PC, one switch, and one server.
- 3. Connect the PC and server through the switch.
- 4. Configure the IP address of the client PC.
- 5. Configure the IP address of the web server.
- 6. Open the server configuration.
- 7. Enable the HTTP service.
- 8. Open the Web Browser on the client PC.
- 9. Enter the server IP address:

TCP Three-Way Handshake

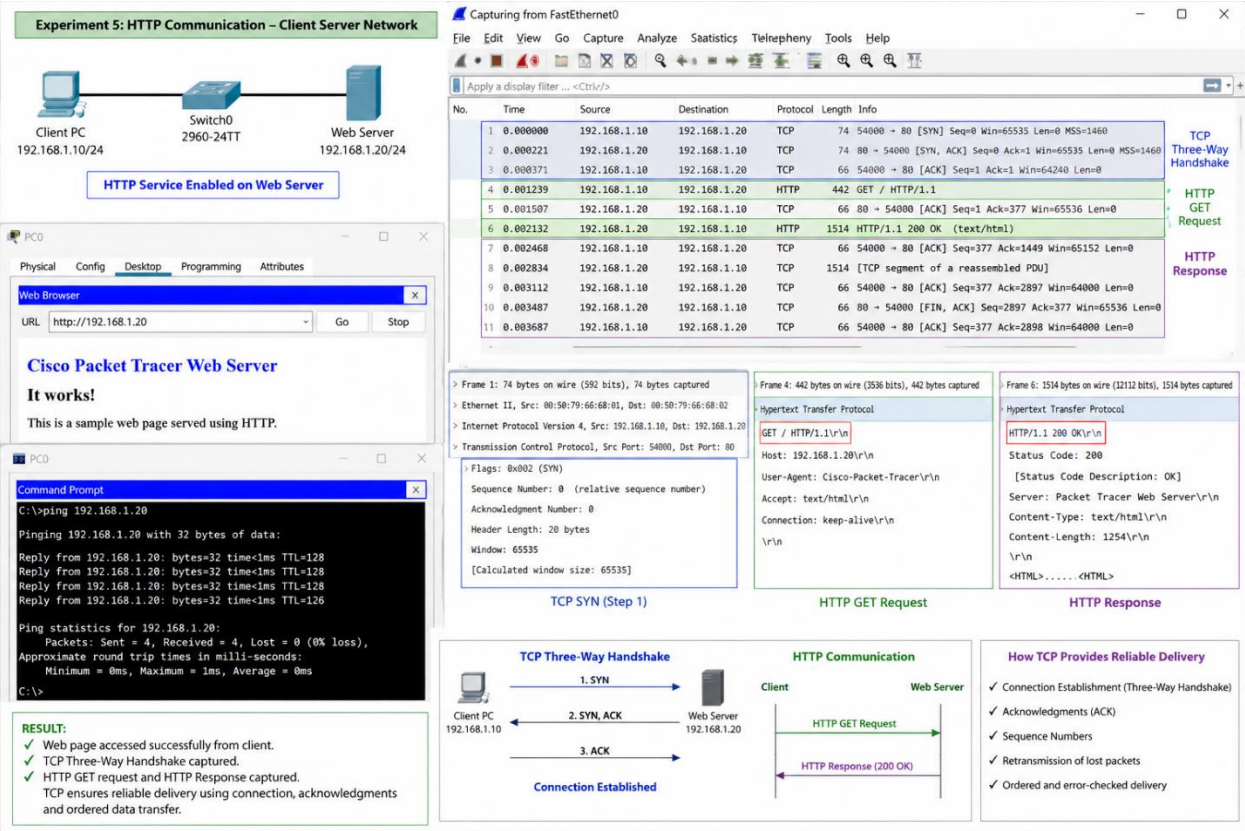
Before HTTP data is transferred, TCP establishes a connection.

Complete Communication Flow:



Result

The client successfully accessed the webpage hosted on the web server. The TCP three-way handshake, HTTP GET request, and HTTP response were successfully analyzed. TCP provided reliable communication through connection establishment, acknowledgments, retransmission, and ordered delivery



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding ; limited or partial integration	Poor understanding ; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

	Understanding of Concepts (25%)	Experimental Design (30%)	Use of Tools (30%)	Analysis of Results (15%)	Documentation (10%)
Q1					
Q2					
Q3					
Q4					
Q5					